BASES ADMITTING SIX- AND EIGHT-DIGIT KAPREKAR FIXED POINTS: A COMPLETE CONGRUENCE CLASSIFICATION

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Abstract. For the Kaprekar transformation K_b (sort the base-b digits in descending and ascending order and subtract) acting on integers with a fixed digit length n, we determine, for $n = 6$ and $n = 8$, exactly which bases $b \geq 2$ admit a fixed point. For $n = 6$: a fixed point exists if and only if $2 \mid b$, or $b \equiv 6 \pmod 7$, or $b \equiv 8 \pmod 9$, or $b \equiv 10 \pmod{15}$. For $n = 8$: if and only if $3 \mid b$, or $b \equiv 10, 12 \pmod{15}$, or $b \equiv 11, 14 \pmod{17}$, or $b \equiv 13 \pmod{21}$, or $b \in \{2, 4, 8\}$. All fixed points are given by explicit families, linear in b, together with a finite exceptional set confined to $b \leq 10$. The proofs are computer-assisted but exact: a channel decomposition of the borrow chain reduces the problem to finitely many linear systems over $\mathbb{Q}$, classified with exact rational arithmetic, and the indeterminate cases are closed by exact two-variable linear programming. We describe the structural transition between the two classifications: the moduli-doubling phenomenon for odd bases persists, while the universal even-base witness of $n = 6$ is replaced at $n = 8$ by a parity-blind $3 \mid b$ family. We also formulate conjectures for $n = 10$.

1. Introduction

Fix a base $b \geq 2$ and a digit length n. For an integer x with exactly n digits in base b (leading digit nonzero), let $\operatorname{desc}(x)$ and $\operatorname{asc}(x)$ denote the integers obtained by sorting the digit string of x (leading zeros permitted after sorting) in descending and ascending order, and set

$$K_{b,n}(x) \;=\; \operatorname{desc}(x) - \operatorname{asc}(x).$$

Kaprekar's celebrated observation [1] is that in base 10 with $n = 4$ every non-repdigit integer reaches the fixed point 6174. The literature on the Kaprekar transformation splits into two natural axes. Along the first axis one fixes the *base* and studies all digit lengths: this is the programme completed for base 10 constants by Prichett, Ludington and Lapenta [6], refined by Iwasaki [9], and developed for general odd and even bases in the two-part work of Kay and Downes-Ward [7, 8]; Ludington [5] showed that every base admits only finitely many Kaprekar constants. Along the second axis one fixes the *digit length* n and varies the base. Here two questions must be distinguished. The classical, stronger question asks for *Kaprekar constants*: a unique fixed point attracting every non-repdigit n-digit integer. Prichett and Hasse [2] determined all bases with a four-digit Kaprekar constant; Lapenta, Ludington and Prichett [3] gave an algorithm for self-producing (= fixed) r-digit g-adic integers. Thakur [4] settled the odd lengths $n = 5, 7, 9$ (with a single base each for $n = 7, 9$), gave partial data for $n = 6, 8$ (Kaprekar constants found only at $b = 13, 17$ for $n = 6$ among $b \leq 39$, and only at $b = 3$ for $n = 8$ among $b \leq 17$), and posed as an open problem exactly the existence question we answer here ("characterization of (B, D) such that $\mathrm{FP}(B, D)$ is non-empty"). The very recent cross-base study of $n = 4$ dynamics in [10] shows the axis remains active. The weaker *existence* question (which bases admit *any* n-digit fixed point) is the natural first layer of that programme, and it

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is the question we settle here for the even lengths $n = 6$ and $n = 8$. The distinction matters: for instance, our scanner shows that for $b \le 60$ five-digit fixed points exist precisely for $b = 2$ and $3 \mid b$, whereas five-digit Kaprekar *constants* are far rarer.

To our knowledge the even lengths $n \ge 6$ have remained open on the second axis. This paper closes the first two cases.

Theorem 1 ($n = 6$)**.** *A base $b \ge 2$ admits a six-digit Kaprekar fixed point if and only if*

$$2 \mid b \quad \vee \quad b \equiv 6 \pmod 7 \quad \vee \quad b \equiv 8 \pmod 9 \quad \vee \quad b \equiv 10 \pmod{15}.$$

All fixed points belong to the four explicit families of Table 1, apart from a finite exceptional set of points confined to bases $b \le 10$ (each such base already being covered by the families). Restricted to odd bases the condition becomes $b \equiv 13$ (14) $\vee$ $b \equiv 17$ (18) $\vee$ $b \equiv 25$ (30).

Theorem 2 ($n = 8$)**.** *A base $b \ge 2$ admits an eight-digit Kaprekar fixed point if and only if*

$$3 \mid b \ \vee\ b \equiv 10, 12 \pmod{15} \ \vee\ b \equiv 11, 14 \pmod{17} \ \vee\ b \equiv 13 \pmod{21} \ \vee\ b \in \{2, 4, 8\}.$$

All fixed points belong to the six explicit families of Table 2, apart from nine exceptional points confined to $b \in \{2, 4, 6, 8, 10\}$; the bases $2, 4, 8$ are the only ones whose fixed points are all exceptional. Restricted to odd bases the condition becomes $b \equiv 3$ (6) $\vee$ $b \equiv 25, 27$ (30) $\vee$ $b \equiv 11, 31$ (34) $\vee$ $b \equiv 13$ (42).

The proofs are computer-assisted in the strong sense: every step is either an elementary identity (the borrow-chain successor formulas of Section 2) or a finite computation carried out in exact rational arithmetic, with the two logical directions treated asymmetrically: every *existence* claim is sealed by directly verifying $K_{b,n}(x) = x$ for witnesses, while every *emptiness* claim uses only necessary linear constraints (Section 3). The complete classification data and the verification code accompany the paper as ancillary files.

Structurally, the two theorems exhibit a striking transition (Section 6): the moduli-doubling phenomenon for odd bases (each family modulus D doubles to $2D$ under the odd-base restriction) persists from $n = 6$ to $n = 8$, but the palindromic witness that makes *every* even base admissible at $n = 6$ has no analogue at $n = 8$; its structural slot is occupied by a parity-blind $3 \mid b$ family, so that infinitely many even bases (e.g. $b \equiv 16 \pmod{30}$ outside the other families) admit no eight-digit fixed point. The asymptotic density of admissible bases is $68/105 \approx 0.648$ for $n = 6$ and $60/119 \approx 0.504$ for $n = 8$, the latter being identical for odd and even bases.

2. Preliminaries and the channel decomposition

Throughout, n is even, $h := n/2$, and $B := b - 1$.

Definition 3. Let x have base-b digits sorted as $a_0 \ge a_1 \ge \cdots \ge a_{n-1}$. The *difference vector* of x is $d(x) = (d_0, \ldots, d_{h-1})$ with $d_i = a_i - a_{n-1-i}$.

Sortedness forces $d_0 \ge d_1 \ge \cdots \ge d_{h-1} \ge 0$. A direct computation gives

$$K_{b,n}(x) \;=\; \sum_{i=0}^{h-1} d_i \left(b^{n-1-i} - b^i\right), \tag{1}$$

so $K_{b,n}$ depends on x only through $d(x)$, and x is a fixed point if and only if x equals the right-hand side of (1) evaluated at $d(x)$. This identity also underlies our ground-truth scanner: enumerating difference vectors costs $O(b^h)$ per base instead of $O(b^n)$.

Resolving the borrow chain in (1) yields the following *channel decomposition*. Let $t \in \{0, \ldots, h-1\}$ be the number of trailing zeros of $d(x)$ and $m := h - t$.

Lemma 4 (successor formulas)**.** *If x is a non-repdigit n-digit integer with difference vector d having t trailing zeros, then the digit string of $K_{b,n}(x)$, from most to least significant, is*

$$\left[\, d_0,\ d_1,\ \ldots,\ d_{m-2},\ d_{m-1}-1,\ \underbrace{B,\ldots,B}_{2t},\ B-d_{m-1},\ \ldots,\ B-d_1,\ B+1-d_0 \,\right].$$

In particular the digit sum of any n-digit fixed point equals $(h+t)B$, and the digit-sum multiples kB with $k=h,\ldots,2h-1$ partition all fixed points into h channels.

Proof. The column differences of desc$-$asc, from most to least significant, are $d_0,\ldots,d_{h-1},-d_{h-1},\ldots,-d_0$. Working from the right: the column $-d_0$ borrows (as $d_0\geq 1$), producing digit $b-d_0$; each subsequent column $-d_i-1$ with $i\geq 1$ is negative, producing $B-d_i$; the $2t$ central columns $0-1$ propagate the borrow producing digit B; the first nonzero column from the centre, $d_{m-1}-1\geq 0$, absorbs the borrow; the remaining columns are unchanged. Summing the resulting digits gives $(h+t)B$. □

For $n=6$ the channels have digit sums $3B, 4B, 5B$; for $n=8$ they are $4B,\ldots,7B$. Empirically (and, after Section 3, provably) only the channels $t=0$ and $t=1$ carry infinite families; channels $t\geq 2$ are overdetermined by t consistency conditions and collapse to finitely many small-base points.

3. Method: cells, exact classification, and closure

Fix n and a channel t, and let $E_1,\ldots,E_n$ denote the n successor expressions of Lemma 4, each linear in $(d_0,\ldots,d_{m-1})$ and B. If x is a fixed point, its multiset of digits *is* $\{E_j\}$; choosing a permutation σ that sorts the expressions into weakly decreasing order therefore places x in the *cell* σ, where the fixed-point condition becomes the linear system

$$E_{\sigma(i)} - E_{\sigma(n-1-i)} \;=\; d_i \qquad (i=0,\ldots,h-1,\ \text{with } d_i := 0 \text{ for } i\geq m),$$

that is, $A_\sigma\, d = B\, c_\sigma + e_\sigma$ with integer data. Every fixed point lies in at least one cell (ties are broken arbitrarily; all cell inequalities are closed), so classifying all cells classifies all fixed points.

Each cell is classified over $\mathbb{Q}$ by row reduction in exact rational arithmetic; since the right-hand side is linear in B, a unique solution has the closed form $d_i = (\alpha_i B+\beta_i)/\Delta$. The cells fall into four classes: unique solution for all B; unique solution at a single B; inconsistent; and underdetermined. For cells unique in B we then impose the full validity system — the sorting inequalities of σ, the digit bounds $0\leq E_j\leq B$, $d_0\geq 1$, the channel condition $d_{m-1}\geq 1$, and monotonicity of d — all of which are linear in B, giving exact thresholds; integrality of the d_i imposes congruences on B modulo Δ. A cell whose inequalities hold for all large B and whose congruences are solvable yields an *infinite family*; a cell whose inequalities hold only in a window $[B_{\min}, B_{\max}]$ is exhausted by direct enumeration.

Two design principles keep the argument airtight.

(i) One-way door. Every existence claim (family members, exceptional points) is finally verified by evaluating (1) and checking $K_{b,n}(x)=x$ directly; every emptiness claim uses only constraints that are provably necessary for a fixed point, all closed under equality. An off-by-one in a constraint can therefore cost efficiency, never correctness.

(ii) Exact closure of underdetermined cells. In an underdetermined cell the solution set is affine in (B,u), where the free parameter u is itself one of the $d_j\in[0,B]$. In every such cell arising for $n\in\{6,8\}$ the free dimension is 1, so the validity region is a polyhedron in the (B,u)-plane. Since the constraints include $B\geq 1$ and $0\leq u\leq B$, the region contains no line; hence, if nonempty, it has a vertex, and vertex enumeration over all constraint pairs (again in exact arithmetic) decides emptiness and, when bounded, computes $\max B$ exactly. Cells fixing a single B (the "finite-B" underdetermined cells) are exhausted by direct enumeration of the free parameters at that base.

Finally, the entire pipeline is anchored to ground truth: an exhaustive difference-vector scan of all bases $b \le 60$ (and, as an independent convention check, a naive full-range scan over all n-digit integers for small bases), against which every symbolic conclusion is compared point-by-point, plus an extension scan ($b \le 150$ for $n = 6$, $b \le 100$ for $n = 8$) confirming the classifications beyond the range used to build them. A further audit re-derives the verdicts of 1200 randomly sampled cells per length by exhaustive enumeration of the descending d-cone at two bases, independently of the linear algebra code.

4. The classification for $n = 6$

Table 1. The four families for $n = 6$ ($B = b - 1$; missing d-components are 0).

Channel	Congruence	d-vector	First members
$t = 1$	$b \equiv 0\ (2)$	$\left(\frac{B+1}{2}, \frac{B+1}{2}\right)$ (palindromic M_b)	$2, 4, 6, \ldots$
$t = 0$	$b \equiv 6\ (7)$	$\left(\frac{5B+3}{7}, \frac{3B-1}{7}, \frac{B+2}{7}\right)$	$6, 13, 20, \ldots$
$t = 0$	$b \equiv 8\ (9)$	$\left(\frac{7B+5}{9}, \frac{5B+1}{9}, \frac{B+2}{9}\right)$	$8, 17, 26, \ldots$
$t = 0$	$b \equiv 10\ (15)$	$\left(\frac{3B+3}{5}, \frac{B}{3}, \frac{B+1}{5}\right)$	$10, 25, 40, \ldots$

The cell census for the main channel $t = 0$ of $n = 6$: of the 720 orderings, 568 have a unique solution for all B (of which exactly 3 survive as infinite families), 140 admit no valid base, and 12 are underdetermined. The twelve underdetermined cells are closed exactly by the linear programming argument of Section 3: eleven have empty validity regions and one is bounded with $\max b \le 60$, inside the exhaustively scanned range. The channel $t = 1$ contains no underdetermined cells and exactly one family, with modulus 2 and residue 0; consequently *no odd base has a fixed point in the* $4B$ *channel*, previously an empirical observation and now a consequence of the classification. Ground truth (52 fixed points for $b \le 60$, digit-sum split $k = 3$:22, $k = 4$:30) matches the families plus exceptional points exactly, as does the extension scan to $b \le 150$.

The density of admissible bases is $408/630 = 68/105$; among odd bases it is $186/630 = 31/105 \approx 0.295$, matching the classical count.

5. The classification for $n = 8$

Table 2. The six families for $n = 8$.

Channel	Congruence	d-vector	First members
$t = 1$	$b \equiv 0\ (3)$	$\left(\frac{2B+2}{3}, \frac{2B-1}{3}, \frac{B+1}{3}\right)$	$3, 6, 9, \ldots$
$t = 0$	$b \equiv 10\ (15)$	$\left(\frac{3B+3}{5}, \frac{B}{3}, \frac{B}{3}, \frac{B+1}{5}\right)$	$10, 25, 40, 55$
$t = 0$	$b \equiv 12\ (15)$	$\left(\frac{13B+7}{15}, \frac{11B-1}{15}, \frac{7B-2}{15}, \frac{B+4}{15}\right)$	$12, 27, 42, 57$
$t = 0$	$b \equiv 11\ (17)$	$\left(\frac{11B+9}{17}, \frac{7B-2}{17}, \frac{5B+1}{17}, \frac{3B+4}{17}\right)$	$11, 28, 45$
$t = 0$	$b \equiv 14\ (17)$	$\left(\frac{15B+9}{17}, \frac{13B+1}{17}, \frac{9B+2}{17}, \frac{B+4}{17}\right)$	$14, 31, 48$
$t = 0$	$b \equiv 13\ (21)$	$\left(\frac{5B+3}{7}, \frac{3B-1}{7}, \frac{B}{3}, \frac{B+2}{7}\right)$	$13, 34, 55$

The nine exceptional points have difference vectors $(1,1,1,1), (1,1,1,0), (1,1,0,0)$ at $b = 2$; $(3,1,1,1), (3,3,1,1), (3,3,3,1)$ at $b = 4$; $(5,5,3,1)$ at $b = 6$; $(7,5,3,1)$ at $b = 8$; and $(9,7,5,1)$ at $b = 10$.

Cell census: channel $t = 0$ has 40320 cells (33344 unique-for-all-B, of which 9 live cells give the five families of Table 2 after deduplication; 505 bounded windows, all exhausted; 5152 inconsistent; 1824 underdetermined). Channel $t = 1$ has 20160 distinct cells and exactly one family; channels $t = 2, 3$ close completely (only the $b = 2$ point, resp. nothing). Of the underdetermined cells, all 1162 finite-B ones fix bases $b \le 60$, covered by the exhaustive scan; the 824 all-B ones all have free dimension one, and the exact LP closure finds 779 empty regions and 45 bounded with $\max b \le 4$, and none unbounded. Hence the classification is complete for every base.

Remark 5 (base ten)**.** The two classical eight-digit fixed points of base 10 (listed, among others, in OEIS [11]) are of different kinds: 63317664 (difference vector $(6, 3, 3, 2)$) is the first member of the $b \equiv 10 \pmod{15}$ family, while 97508421 ($(9, 7, 5, 1)$) is exceptional: it belongs to no infinite family.

The density of admissible bases is $900/1785 = 60/119 \approx 0.504$; because every family modulus is odd, the density is the same among odd and among even bases.

6. Structural comparison and the parity mechanism

As an independent check, Thakur's data [4] on which bases carry a full attracting Kaprekar constant (a strictly stronger property than mere existence of a fixed point) sit inside our classification, as they must: his $n = 6$ constants at $b = 13, 17$ are the smallest members of our $b \equiv 13$ (14) and $b \equiv 17$ (18) families, and his $n = 8$ constant at $b = 3$ is the smallest member of our $3 \mid b$ family.

Table 3. Structural comparison.

	$n = 6$	$n = 8$
channels (live)	$3B, 4B$ (of $3B, 4B, 5B$)	$4B, 5B$ (of $4B, \ldots, 7B$)
even bases	all admissible	not all ($16, 20, 22, \ldots$ empty)
$t = 1$ witness divisor	2 (M_b, palindromic)	3
families / moduli	4 / $\{2, 7, 9, 15\}$	6 / $\{3, 15, 15, 17, 17, 21\}$
odd-base moduli	$14, 18, 30$ (doubled)	$6, 30, 34, 42$ (doubled)
density (all / odd)	$68/105$ / $31/105$	$60/119$ / $60/119$

This table has two sources. In channel t the system has h equations and $m = h-t$ unknowns, so $t = 0$ is square (many families), $t = 1$ carries exactly one consistency condition (one distinguished family), and $t \ge 2$ is overdetermined (finite collapse). On top of that, the single $t = 1$ family has divisor $h - 1$: for $n = 6$ this is 2, forcing $2 \mid b$ and creating the even/odd asymmetry, while for $n = 8$ it is 3, which is parity-blind. Since all $t = 0$ moduli are odd in both cases, the parity wall of $n = 6$ is thereby explained as an artefact of $h - 1 = 2$, not a general feature.

Conjecture 6 ($n = 10$)**.** *For $n = 10$ there are five channels with digit sums $5B, \ldots, 9B$, of which only $5B$ and $6B$ carry infinite families; the distinguished $t = 1$ family has divisor $h - 1 = 4$, so $b \equiv 0 \pmod 4$, restoring a (weak) even-base advantage; all $t = 0$ moduli are odd, and the family moduli again double under the odd-base restriction. The set of admissible odd bases has density strictly smaller than at $n = 8$.*

The machinery of Section 3 applies verbatim to $n = 10$ (the $t = 0$ channel has $10!$ cells, which is within reach of the exact pipeline; a provable reduction to riffle-shuffle cells, using the fact that the internal orders of the two blocks of successor expressions are forced, cuts this to a few thousand).

7. Reproducibility

All computations use exact integer and rational arithmetic (no floating point). The ancillary files contain: the core scanner and the general even-n pipeline (`core.py`, `genpipe.py`, and the $n = 8$ phase scripts), the ground-truth data ($b \leq 60$), the complete cell classifications, and the audit scripts. A single command reruns each stage; the $n = 6$ hardening run reproduces the historical cell census 568/140/12 found in the author's original computation, an independent cross-validation of both implementations.

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